THE NEED FOR AUTONOMOUS SYSTEMS IN DISASTERS

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UAVS IN DISASTER RESPONSE

- Mapping with LiDAR, GNSS, and photogrammetry.
- Thermal detection for finding trapped survivors.
- UAV-BS units restore communications when towers fail.

GROUND-BASED ROBOTICS

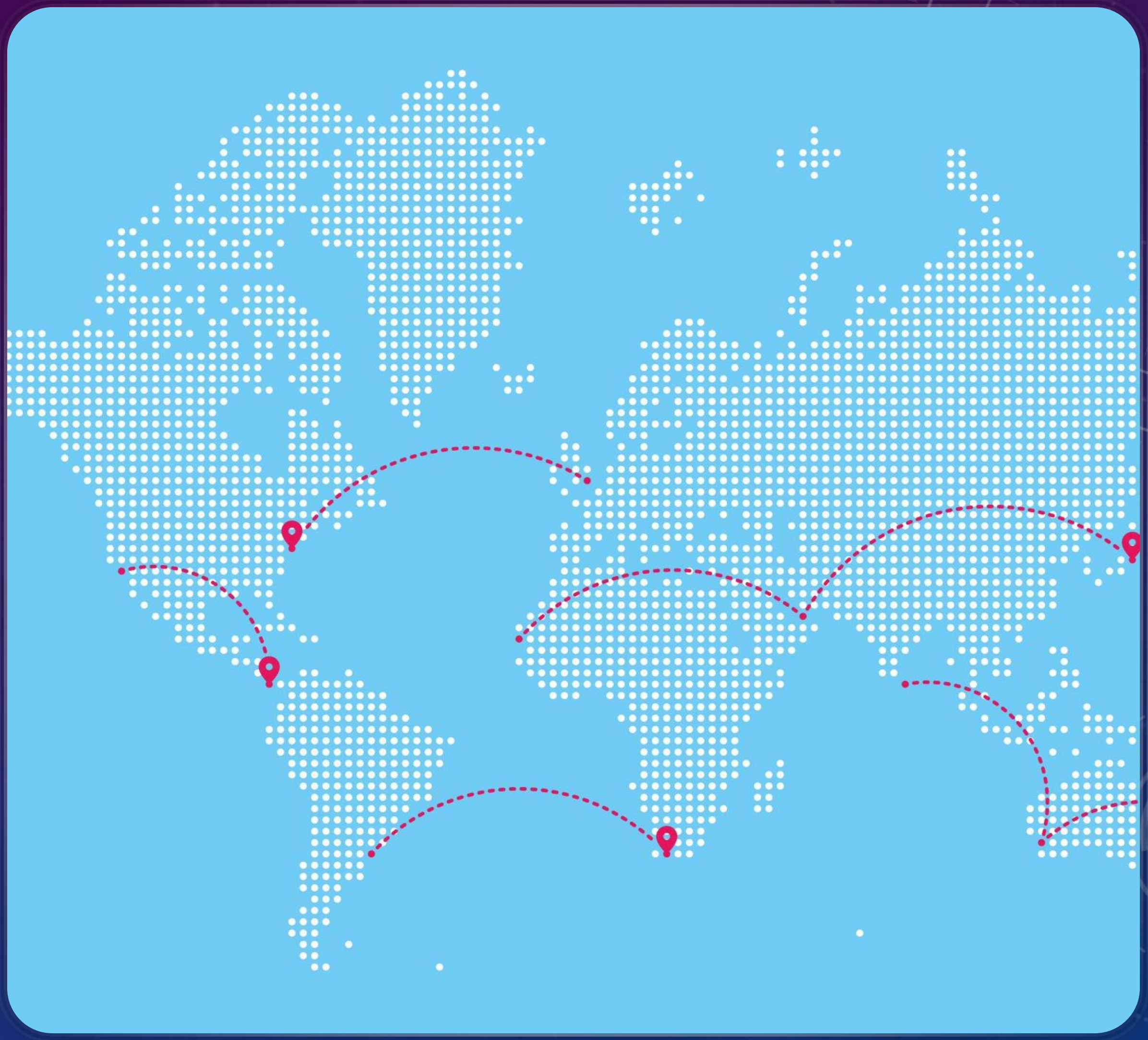
Inspection in unsafe or collapsed areas.

Hazard sampling (toxic chemicals, radiation).

Advanced mobility & tele-autonomy.

REAL LIFE EXAMPLES

.**Australian Bushfires**: Thermal cameras & mapping improved evacuation routes and wildlife rescue.
Nepal Earthquake: UAV mapping identified worst-hit areas and guided aid



PROS/ CONS

Faster response, reduced risk to humans, precise mapping.

Limited battery life, environmental interference, GPS-denied navigation, high costs, cybersecurity risks

Conclusion: Indispensable but not flawless — must be integrated with human efforts.